

Extended Body Control System - Extended High-Level Design

Document ID: HLD-BCM-100-001 Version: 1.0 Status: Synthetic Demonstration Document

1. Purpose and Scope

This synthetic AUTOSAR-style HLD describes the Extended Body Control System. It is intentionally long to test page-wise ingestion, chunking, semantic retrieval, structured architecture extraction, validation, graph construction, revision tooling, and evidence-grounded question answering. The architecture facts are deterministic; surrounding engineering prose provides realistic document length without introducing confidential OEM data.

2. Architecture Summary

Component	Responsibility	Key Interface
LightSwitch	Captures driver exterior-light requests.	LightRequestInterface
BodyControlManager	Coordinates body-control functions and processes light requests.	BodyCommandInterface
ExteriorLightController	Controls exterior lighting based on body commands.	LightStatusInterface
DoorStateManager	Aggregates door-state information for body functions.	DoorStateInterface
DoorSensorAdapter	Adapts raw door sensor states into architecture-level information.	DoorSensorInterface
InteriorLightController	Controls interior lighting using door state and body commands.	InteriorLightStatusInterface
BodyDiagnostics	Monitors body-control components and publishes diagnostic state.	BodyDiagnosticInterface
BodyGateway	Exposes selected body status to external vehicle domains.	BodyGatewayInterface

2.1 Document Processing Guidance

The document should be processed incrementally by page. Native text extraction is preferred; OCR is a fallback for image-only pages. Retrieval should operate on chunks rather than passing the complete document to the language model. Architecture extraction should normalize equivalent labels before mapping facts into a common architecture model.

3. Component Definitions

Component Name	Description
LightSwitch	Captures driver exterior-light requests.
BodyControlManager	Coordinates body-control functions and processes light requests.
ExteriorLightController	Controls exterior lighting based on body commands.
DoorStateManager	Aggregates door-state information for body functions.
DoorSensorAdapter	Adapts raw door sensor states into architecture-level information.
InteriorLightController	Controls interior lighting using door state and body commands.
BodyDiagnostics	Monitors body-control components and publishes diagnostic state.
BodyGateway	Exposes selected body status to external vehicle domains.

4. Interface Definitions

Interface Name	Provider	Consumer
LightRequestInterface	LightSwitch	BodyControlManager
BodyCommandInterface	BodyControlManager	ExteriorLightController
LightStatusInterface	ExteriorLightController	BodyDiagnostics
DoorStateInterface	DoorStateManager	InteriorLightController
DoorSensorInterface	DoorSensorAdapter	DoorStateManager
InteriorLightStatusInterface	InteriorLightController	BodyControlManager
BodyDiagnosticInterface	BodyDiagnostics	BodyGateway
BodyGatewayInterface	BodyGateway	ExternalDomain

5. Port Definitions

Component	Port Name	Port Type	Interface
LightSwitch	LightSwitchOut	P-Port	LightRequestInterface
BodyControlManager	BodyControlManagerOut	P-Port	BodyCommandInterface
ExteriorLightController	ExteriorLightControllerOut	P-Port	LightStatusInterface
DoorStateManager	DoorStateManagerOut	P-Port	DoorStateInterface
DoorSensorAdapter	DoorSensorAdapterOut	P-Port	DoorSensorInterface
InteriorLightController	InteriorLightControllerOut	P-Port	InteriorLightStatusInterface
BodyDiagnostics	BodyDiagnosticsOut	P-Port	BodyDiagnosticInterface
BodyGateway	BodyGatewayOut	P-Port	BodyGatewayInterface
BodyControlManager	LightRequestIn	R-Port	LightRequestInterface
ExteriorLightController	BodyCommandIn	R-Port	BodyCommandInterface
DoorStateManager	DoorSensorIn	R-Port	DoorSensorInterface
InteriorLightController	DoorStateIn	R-Port	DoorStateInterface
BodyDiagnostics	LightStatusIn	R-Port	LightStatusInterface
BodyGateway	DiagnosticIn	R-Port	BodyDiagnosticInterface

6. Signal Definitions

Signal Name	Interface	Unit
LightRequest	LightRequestInterface	enum
BodyCommand	BodyCommandInterface	enum
LightStatus	LightStatusInterface	enum
DoorState	DoorStateInterface	enum
BodyDiagnosticStatus	BodyDiagnosticInterface	enum

7. Component Dependencies

Dependent Component	Required Component
BodyControlManager	LightSwitch
BodyControlManager	DoorStateManager
ExteriorLightController	BodyControlManager
DoorStateManager	DoorSensorAdapter
InteriorLightController	DoorStateManager
InteriorLightController	BodyControlManager
BodyDiagnostics	BodyControlManager
BodyDiagnostics	ExteriorLightController
BodyGateway	BodyControlManager
BodyGateway	BodyDiagnostics

8. Functional Flows

Flow ID	Architecture Flow
FLOW-01	LightSwitch -> BodyControlManager -> ExteriorLightController
FLOW-02	DoorSensorAdapter -> DoorStateManager -> InteriorLightController
FLOW-03	BodyControlManager -> BodyDiagnostics -> BodyGateway

8.1 Traceability Notes

Each flow is derived from explicit component, interface, port, signal, and dependency relationships. Generated analysis must remain traceable to the source pages and must not be treated as automatic architecture approval.

6. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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7. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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8. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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9. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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10. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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11. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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12. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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13. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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14. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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15. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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16. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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17. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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18. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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19. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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20. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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21. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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22. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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23. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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24. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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25. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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26. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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27. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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28. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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29. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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30. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

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31. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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32. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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33. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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34. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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35. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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36. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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37. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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38. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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39. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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40. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

This page intentionally contains enough technical prose to create multiple retrieval chunks across the full 100-page document. A scalable implementation should index chunks once, retrieve only the most relevant evidence for each question, and avoid sending all pages to the language model. Page metadata should be retained so every answer can be traced back to its source.

41. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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42. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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43. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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44. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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45. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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46. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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47. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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48. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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49. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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50. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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51. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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52. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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53. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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54. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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55. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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56. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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57. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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58. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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59. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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60. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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61. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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62. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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63. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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64. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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65. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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66. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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67. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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68. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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69. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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70. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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71. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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72. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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73. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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74. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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75. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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76. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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77. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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78. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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79. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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80. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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81. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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82. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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83. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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84. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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85. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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86. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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87. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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88. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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89. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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90. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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91. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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92. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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93. Engineering Detail - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains.

Primary interface. BodyGatewayInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyGateway participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyGateway to BodyGatewayInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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94. Engineering Detail - LightSwitch

Responsibility. Captures driver exterior-light requests.

Primary interface. LightRequestInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

LightSwitch participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace LightSwitch to LightRequestInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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95. Engineering Detail - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests.

Primary interface. BodyCommandInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyControlManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyControlManager to BodyCommandInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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96. Engineering Detail - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands.

Primary interface. LightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

ExteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace ExteriorLightController to LightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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97. Engineering Detail - DoorStateManager

Responsibility. Aggregates door-state information for body functions.

Primary interface. DoorStateInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorStateManager participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorStateManager to DoorStateInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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98. Engineering Detail - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information.

Primary interface. DoorSensorInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

DoorSensorAdapter participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace DoorSensorAdapter to DoorSensorInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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99. Engineering Detail - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands.

Primary interface. InteriorLightStatusInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

InteriorLightController participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace InteriorLightController to InteriorLightStatusInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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100. Engineering Detail - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state.

Primary interface. BodyDiagnosticInterface. The interface is reviewed for ownership, provider/consumer direction, port consistency, signal traceability and change impact.

Design Review Context

BodyDiagnostics participates in the architecture described by this HLD. Review activities include checking interface contracts, dependency direction, initialization assumptions, diagnostic visibility, failure propagation, data ownership, traceability, and downstream impact. The purpose of this section is to provide realistic surrounding engineering context while preserving the same deterministic architecture facts.

Review Item	Expected Treatment
Architecture ownership	Trace BodyDiagnostics to BodyDiagnosticInterface and related dependencies.
Evidence grounding	Answers must cite retrieved HLD evidence.
Change impact	Compare component, interface, port, signal and dependency changes.
Human oversight	Engineer reviews generated findings before design decisions.

Scalability Test Narrative

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